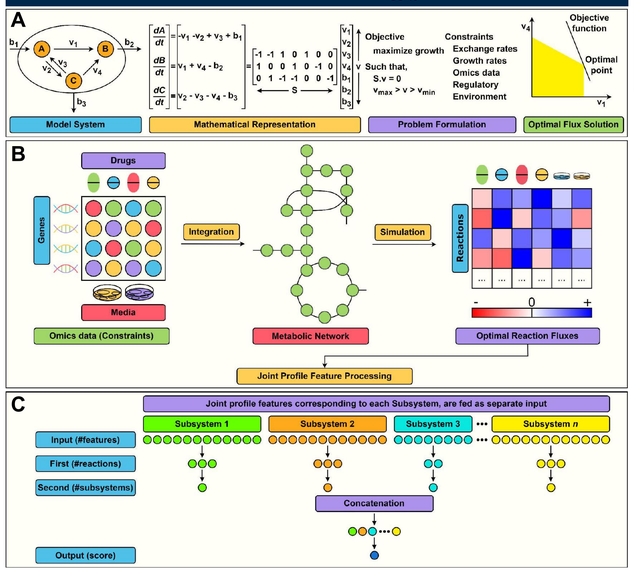
A mechanistic neural network model optimizing both the potency and toxicity of combination therapies

Harkirat Singh Arora[*,1], Katherine Lev[*,2], Aaron Robida[3], Sriram Chandrasekaran[1,2]
[1]Department of Biomedical Engineering, [2]Program in Chemical Biology, [3]Center for Chemical Genomics, University of Michigan
{hsarora, katlev, bida, csriram} @ umich.edu

Background and Motivation

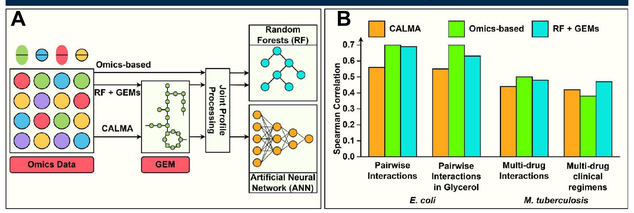
- Antibiotic resistance is becoming a major global public health concern, with current treatments becoming ineffective, and no newer treatments available
- The World Health Organization warns of a potential future where **drug-resistant infections could annually claim 10 million lives by mid-century [1]**
- **Drug combination therapy** is a promising solution for antibiotic resistance but **current strategies revolves around empirical methods**, leading to **suboptimal potency, promote resistance, and pose high toxicity risks**
- Moreover, the high costs associated with experimental testing of numerous possible combinations underscore the **need for a fast and efficient data-driven strategy to facilitate the innovation of combinational therapies**
- We propose an approach, **CALMA (Combining Antibiotics by Leveraging Metabolism-inspired Artificial neural networks)**, which
  - predicts multi-way drug interactions in *E. coli* and *M. tuberculosis*
  - accounts for drug toxicity profiles to design safer treatments
  - provides insights into mechanisms of action of drug combinations
- We envision that our approach will streamline the early-stage drug discovery process, allowing researchers to eliminate less-promising candidates quickly

Methodology



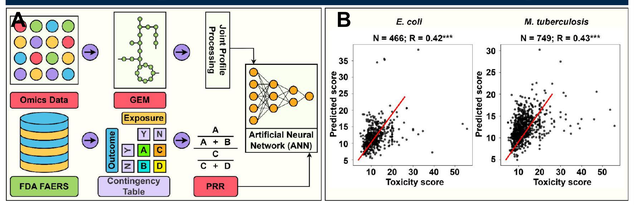
Schematic. (A) Genome-scale Metabolic Models (GEMs) are computational representations of an organism's metabolic network (model system) by annotating an array of reactions and metabolites (mathematical representation). GEMs help to optimize a certain objective function and integrate different constraints (problem formulation) to simulate reaction fluxes (optimal flux solution). Adapted from [2,3]. **(B)** Omics data for different drug and media conditions are integrated as constraints to the metabolic network for simulating reaction fluxes corresponding to these conditions, followed by joint profile processing to construct combination profiles. **(C)** Joint profile features corresponding to the same subsystem are provided as the same input to the metabolism-inspired Artificial Neural Network (ANN) model to predict drug combination potency and toxicity scores.

Benchmarking



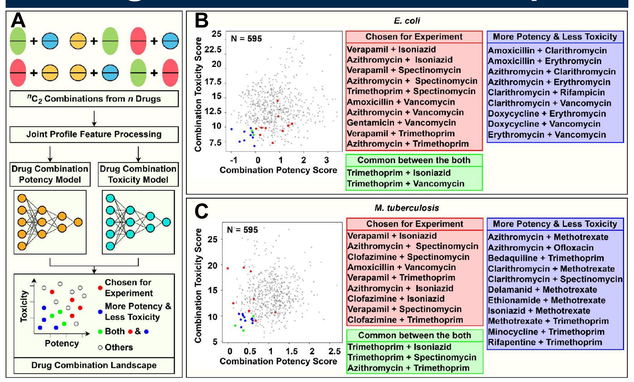
Benchmarking. CALMA can predict multi-way drug combination potency scores in *E. coli* and *M. tuberculosis*. **(A)** Schematic representation of benchmarking of CALMA with difference approaches: Omics-based and RF+GEMs. **(B)** Bar chart visualizing Spearman rank correlation between experimental and predicted scores in several datasets for different approaches: CALMA, Omics-based (INDIGO [4], INDIGO-MTB [5]), Random Forests (RF)+GEMs (CARAMeL [6]).

Drug Combination Toxicity



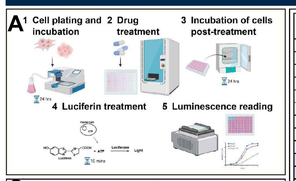
Drug Combination Toxicity. Safety is an important concern in designing drug combination therapies for researchers. **(A)** Schematic representation of training CALMA to predict drug combination toxicity scores. FDA Adverse Event Reporting System (FAERS) database [7] providing information on drug combinations and side effects, with their corresponding toxicity score: Proportional Reporting Ratio (PRR). **(B)** Shown are the scatter plots for *E. coli* and *M. tuberculosis* models in predicting drug combination toxicity with the number of interactions in the dataset (N), and Spearman rank correlation (R). ***p < 0.001

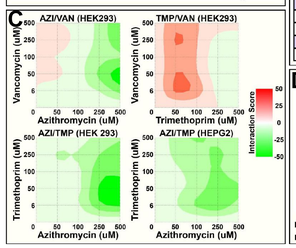
Drug Combination Landscape



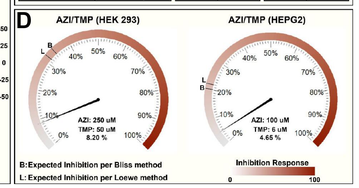
Drug Combination Landscape. The prediction of drug combination potency and toxicity scores with significant accuracy by CALMA helps to plot the drug combination landscape with Combination Interaction Score (CIS) along x-axis, and Combination Toxicity Score (CTS) along y-axis. **(A)** Schematic overview of generation of drug combination landscape. **(B)** Drug Combination Landscape for pairwise combinations in *E. coli* and **(C)** *M. tuberculosis*. Few combinations (in red and green) are chosen for experimental testing for their safety *in vitro*.

Experimental Validation for Safety



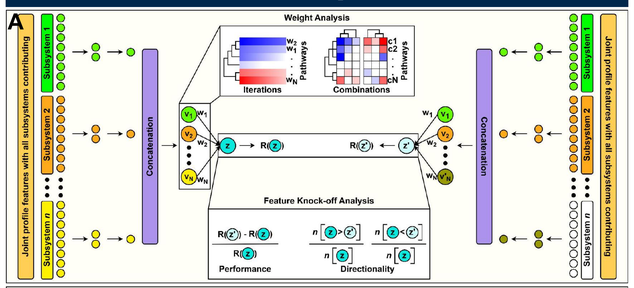


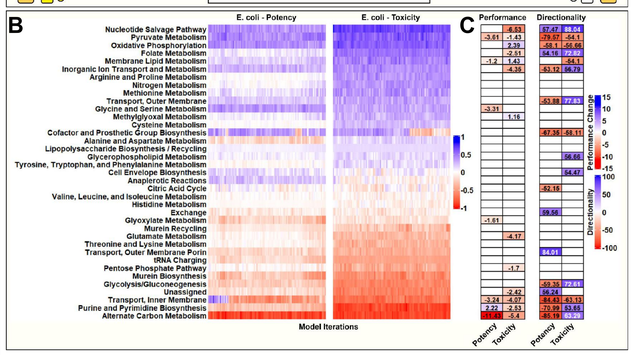
B	Drug Combination		HEK293 (Kidney)		HEPG2 (Liver)	
			Bliss	Loewe	Bliss	Loewe
	Verapamil	Isoniazid	-5.53	-5.77	-4.83	-5.01
	Azithromycin	Isoniazid	-9.78	-10.06	-6.48	-7.35
	Trimethoprim	Isoniazid	-9.08	-9.64	-8.79	-9.03
	Clofazimine	Isoniazid	-7.19	-7.41	-9.84	-10.16
	Verapamil	Spectinomycin	-10.89	-6.77	-7.35	-7.17
	Azithromycin	Spectinomycin	-7.17	-3.89	-3.55	-3.96
	Trimethoprim	Spectinomycin	-4.21	1.75	-5.13	-5.00
	Clofazimine	Spectinomycin	2.83	6.65	-6.80	-6.40
	Amoxicillin	Vancomycin	-7.96	-6.8	-5.83	-8.01
	Azithromycin	Vancomycin	-12.17	-11.65	-7.81	-7.98
	Trimethoprim	Vancomycin	7.38	6.72	-6.48	-5.24
	Gentamicin	Vancomycin	-9.59	-8.23	-5.19	-7.96
	Verapamil	Trimethoprim	-7.30	-4.71	-12.00	-2.33
	Azithromycin	Trimethoprim	-17.67	-16.93	-23.07	-16.66
	Clofazimine	Trimethoprim	-4.31	-0.05	-10.29	-1.97



Experimental Validation. A few selected combinations are tested for their safety in human cells. **(A)** Flowchart for CellTiter-Glo Cell Viability Assay used for combination cytotoxicity experiments in mammalian HEK293 (Kidney) and HEPG2 (Liver) cells. **(B)** Information about cell viability in the presence of combination treatment and individual drug treatments is used to compute the Bliss and Loewe interaction scores using Synergy Finder (lower score = safer). Interesting combinations are highlighted in purple. **(C)** Interaction scores vs Dose Heatmaps for a few interesting combinations. **(D)** Expected vs Actual Inhibition response for Azithromycin and Trimethoprim combination in both cell lines.

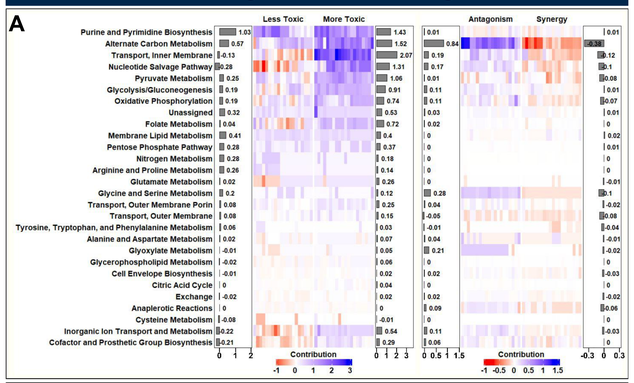
Model Interpretation



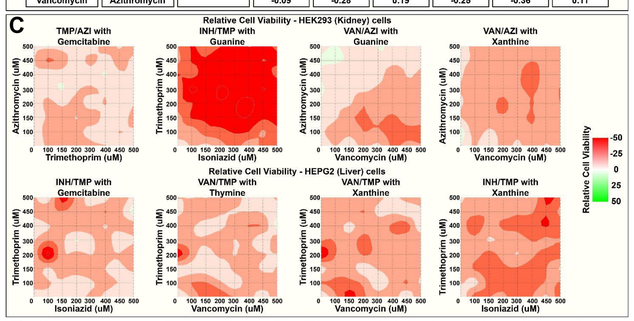


Model Interpretation. The CALMA model is interpreted to understand the mechanisms of action of drug combinations for potency and toxicity. **(A)** Model weights and performance are analyzed based on the impact of different pathways to quantify the importance. **(B)** The heatmap visualizes the normalized weights given by the both potency and toxicity models to different pathways in *E. coli*. **(C)** The heatmap to understand the impact of knocking off a particular pathway on the performance and directionality of predictions in *E. coli*.

Pathway-level Analysis



B	Drug 1 (D1)	Drug 2 (D2)	Metabolite (N)	HEK293 (Kidney)			HEPG2 (Liver)		
				D1 + D2 + N	D1 + D2	Difference	D1 + D2 + N	D1 + D2	Difference
	Vancomycin	Trimethoprim	Gemcitabine	0.39	0.22	0.17	0.17	-0.02	0.19
	Isoniazid	Trimethoprim		-0.06	-0.08	0.02	0.19	0.07	0.12
	Vancomycin	Azithromycin		-0.10	-0.25	0.15	-0.28	-0.39	0.11
	Trimethoprim	Azithromycin		-0.09	-0.21	0.12	-0.18	-0.15	-0.03
	Vancomycin	Trimethoprim	Guanine	0.30	0.22	0.08	0.15	-0.02	0.17
	Isoniazid	Trimethoprim		0.08	-0.08	0.16	0.05	0.07	-0.02
	Vancomycin	Azithromycin		-0.11	-0.28	0.17	-0.32	-0.36	0.04
	Vancomycin	Trimethoprim	Thymine	0.46	0.24	0.22	0.30	0.12	0.18
	Isoniazid	Trimethoprim		-0.15	0.06	-0.21	-0.09	-0.14	0.05
	Vancomycin	Trimethoprim	Xanthine	0.51	0.24	0.27	0.31	0.12	0.19
	Isoniazid	Trimethoprim		-0.09	0.06	-0.16	-0.08	-0.14	0.06
	Vancomycin	Azithromycin		-0.09	-0.28	0.19	-0.26	-0.36	0.11



Pathway-level Analysis. (A) The weighted contributions from different pathways for the top and bottom 20 combinations for potency and toxicity for E. coli, shown as cluster heatmap, along with mean contribution of the individual pathways to the final prediction scores as bar charts. Nucleotide Salvage Pathway is the top pathway governing the drug combination toxicity outcomes. **(B)** Comparison of bliss synergy scores of different drug combinations for human cell lines (HEK293 and HEPG2) in presence and absence of nucleotide metabolites. **(C)** Relative Cell Viability Heatmaps for a few interesting combinations in presence and absence of nucleotide metabolites.

Conclusion and Future Directions

- We expect that the proposed computational pipeline, CALMA will become a valuable instrument in designing tailored combination therapies, contributing to the global effort to curtail and mitigate drug resistance.
- Beyond bacterial infections, CALMA has the potential to be extended to other microbial species such as fungi, and other disease areas such as cancer.

References

1. WHO report on AMR, 2019
2. Kauffman et al., *COBIOT*, 2003
3. Sahu et al., *CSBJ*, 2021
4. Chandrasekaran et al., *Mol. Syst. Biol.*, 2016
5. Ma et al., *mBio*, 2019
6. Chung et al., *PNAS Nexus*, 2022
7. Tatonetti et al., *Sci. Transl. Med.*, 2012
8. Zheng et al., *Gen. Prot. Bio.*, 2022

Acknowledgements

Chandrasekaran Lab
- Special thanks to Carolina Chung

MICHIGAN ENGINEERING
MEDICAL SCHOOL
NIH National Institutes of Health

Scan to learn more about our Lab